

Homework 3 Solutions

Definitions

Define the following terms/concepts

- a.) Sampling Error - the difference between a sample estimate and the true population parameter it is estimating, which arises because we observe only a subset of the population
- b.) Law of Large Numbers - A theorem in probability and statistics which states that the relative frequency of an event will tend to “approach” (in some sense) the probability of an event as the number of observations increases
- c.) Sample Space - the collection of all possible outcomes of a random phenomenon or experiment
- d.) Event - A set of outcomes of a random phenomenon or experiment - that is, any subset of the sample space
- e.) Union - the combined outcomes of two events
- f.) Intersection - the overlapping outcomes of two events
- g.) Random Trial - a process or experiment that can be repeated with a well-defined set of possible outcomes.
- h.) Random Variable - a function that assigns a numerical value to each outcome in the sample space
- i.) Probability distribution - a function that gives the probabilities of different possible outcomes of a random variable
- j.) Population distribution - is the probability distribution for a single observation of a random variable
- k.) Continuous random variable - A random variable with an uncountable number of outcomes
- l.) Discrete random variable - A random variable with a countable number of outcomes
- m.) Probability Mass Function - A function which assigns a probability to each outcome of a discrete random variable
- n.) Probability Density Function - A function which assigns probability density to different outcomes of a continuous random variable

Simple probability

(1) A fair coin is thrown 3 times. What is the probability that at least one heads is observed?

$$P(\text{At least one heads}) = 1 - P(\text{No Heads})$$

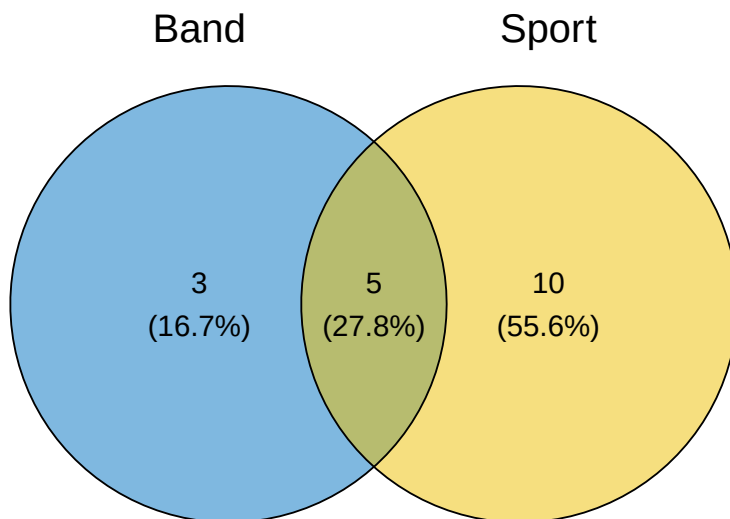
The sample space of this event is

$$\mathbb{S} = \left[\begin{array}{l} \text{No heads} = (T, T, T) \\ 1 \text{ Heads} = (H, T, T), (T, H, T), (T, T, H) \\ 2 \text{ Heads} = (H, H, T), (H, T, H), (T, H, H) \\ \text{All Heads} = (H, H, H) \end{array} \right]$$

The sample space consists of 8 possible outcomes and there is only one outcome with “No Heads”.

$$P(\text{At least one heads}) = 1 - P(\text{No Heads}) = 1 - \frac{1}{8} = \frac{7}{8}$$

(2) Suppose that in a class of 30 students, 8 students are in band, 15 students play a sport, and 5 students are both in band and play a sport. Let A be the event that a student is in band and let B be the event that a student plays a sport. Create a Venn diagram that models this situation. What is the probability that a student is in the band or plays a sport?



Since these events are non-disjoint, we must use the formula for a union of two non-disjoint events:

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= \frac{8}{30} + \frac{15}{30} - \frac{5}{30} = \frac{18}{30} \end{aligned}$$

(3) A standard deck of 52 cards contains four suits (Hearts, Diamonds, Clubs, and Spades). Each suit contains 13 cards (9 numbered cards and 3 face cards, and an Ace).

- Suppose a single card is drawn at random, what is the probability that it is a numbered card?

$$P(\text{Card is numbered}) = \frac{\text{Number of ways to get numbered card}}{\text{Total number of cards}} = \frac{9 \cdot 4}{52}$$

- What is the probability that the selected card is the Ace of Spades (assume independence where necessary)?

Let A be the event that a card drawn at random is an Ace and B be the event that a card drawn at random is a Spades. We are interested in the $P(A \cap B)$ and we are told to assume independence. Therefore

$$P(A \cap B) = P(A) \cdot P(B) = \frac{4}{52} \cdot \frac{13}{52} = \frac{1}{52}$$

(4) An urn contains 5 green balls, 7 red balls, and 4 yellow balls. Two balls are selected at random with replacement. What is the probability that first ball is yellow and the second ball is red?

Since the balls are selected with replacement the two events are independent. Therefore, the intersection of the two events: A first ball yellow and B second ball is red is given by

$$P(A \cap B) = \frac{4}{16} \cdot \frac{7}{16} = \frac{7}{64}$$

(5) The probability that it rains R on a given day in Moscow, Idaho is $P(R) = 0.1$. The probability that it is Windy W is $P(W) = 0.2$, and the probability that it is cloudy C is $P(C) = 0.7$. Assume that these three types of weather events are independent. What is the probability that a given day is Rainy, Cloudy, and Windy?

We are told the events are independent and we wish to find the intersection of R , W and C :

$$P(R \cap W \cap C) = (0.1)(0.2)(0.7) = 0.014$$

Sampling Distributions, Means, and Variances

(6) A random variable X has the following probability distribution

X	P(X)
1	0.25
2	0.25
3	0.25
4	0.25

- Derive the sampling distribution for the mean of $n = 2$ observations of X

The sample space for a mean of two observations of X is given below.

Observation 1	Observation 2	$\bar{X}$
1	1	1.0
2	1	1.5
3	1	2.0
4	1	2.5
1	2	1.5
2	2	2.0
3	2	2.5
4	2	3.0
1	3	2.0
2	3	2.5
3	3	3.0
4	3	3.5
1	4	2.5
2	4	3.0
3	4	3.5
4	4	4.0

The sampling distribution of $\bar{X}$ is given by:

$\bar{X}$	$P(\bar{X})$
1.0	$1(0.25^2) = 0.0625$
1.5	$2(0.25^2) = 0.125$
2.0	$3(0.25^2) = 0.1875$
2.5	$4(0.25^2) = 0.25$
3.0	$3(0.25^2) = 0.1875$
3.5	$2(0.25^2) = 0.125$
4.0	$1(0.25^2) = 0.0625$

- Compute the mean $\mu_{\bar{X}}$ and standard deviation $\sigma_{\bar{X}}$ of the sampling distribution of $\bar{X}$

Using the sampling distribution of $\bar{X}$ derived above:

$$\begin{aligned}
 \mu_{\bar{X}} &= \sum_{\bar{x}} \bar{x}P(\bar{x}) \\
 &= 1(0.0625) + 1.5(0.125) + \cdots + 3.5(0.125) + 4(0.0625) \\
 &= 2.5 \\
 \sigma_{\bar{X}} &= \sqrt{\sum_{\bar{x}} (\bar{x} - \mu_{\bar{X}})^2 P(\bar{x})} \\
 &= \sqrt{(1 - 2.5)^2(0.0625) + (1.5 - 2.5)^2(0.125) + \cdots + (3.5 - 2.5)^2(0.125) + (4 - 2.5)^2(0.0625)} \\
 &= \sqrt{0.625} \\
 &= 0.791
 \end{aligned}$$

Note that this agrees with the population values for X itself: $\mu = 2.5$ and $\sigma = \sqrt{1.25} = 1.118$, and the standard deviation of the sampling distribution is $\sigma/\sqrt{n} = 1.118/\sqrt{2} = 0.791$.

(7) A random variable Y has two possible outcomes. The outcome $Y = 1$ occurs with probability $p = 0.23$. Derive the sampling distribution for the proportion of outcomes with $Y = 1$ in a sample of $n = 3$ observations of Y .

The population distribution of Y is given by:

Y	$P(Y)$
0	0.77
1	0.23

The sample space for the sample proportion of 3 observations of Y is given below.

Observation 1	Observation 2	Observation 3	$\hat{p}$
0	0	0	0.00
1	0	0	0.33
0	1	0	0.33
1	1	0	0.67
0	0	1	0.33
1	0	1	0.67
0	1	1	0.67
1	1	1	1.00

The sampling distribution of $\hat{p}$ is given by:

$\hat{p}$	$P(\hat{p})$
0.00	$1(0.77^3) = 0.457$
0.33	$3(0.23 \times 0.77^2) = 0.409$
0.67	$3(0.23^2 \times 0.77) = 0.122$
1.00	$1(0.23^3) = 0.012$

(8) Consider a survey of tree volume. Suppose that in the population of trees 10% have a volume of 10 cubic feet, 20% have a volume of 20 cubic feet, 30% have a volume of 30 cubic feet, and 40% have a volume of 40 cubic feet. Let X denote the volume of a randomly selected tree, assuming that every tree has an equal chance of being selected. The probability distribution of X is given in the table below.

X	$P(X)$
10	0.1
20	0.2
30	0.3
40	0.4

- Confirm that X has a mean of $\mu = 30$ cubic feet and a standard deviation of $\sigma = 10$ cubic feet by computing them yourself.

$$\begin{aligned}\mu &= \sum_x xP(x) \\ &= 10(0.1) + 20(0.2) + 30(0.3) + 40(0.4) \\ &= 30\end{aligned}$$

$$\begin{aligned}\sigma &= \sqrt{\sum_x (x - \mu)^2 P(x)} \\ &= \sqrt{(10 - 30)^2(0.1) + (20 - 30)^2(0.2) + (30 - 30)^2(0.3) + (40 - 30)^2(0.4)} \\ &= \sqrt{100} \\ &= 10\end{aligned}$$

- Derive the sampling distribution for the mean tree volume in a sample of $n = 2$ trees.

The sample space for the sample mean of 2 observations of tree volume is given below.

Observation 1	Observation 2	$\bar{X}$
10	10	10
20	10	15
30	10	20
40	10	25
10	20	15
20	20	20
30	20	25
40	20	30
10	30	20
20	30	25
30	30	30
40	30	35
10	40	25
20	40	30
30	40	35
40	40	40

The sampling distribution of $\bar{X}$ is given by:

$\bar{X}$	$P(\bar{X})$
10	$(0.1)^2 = 0.01$
15	$2(0.1 \times 0.2) = 0.04$
20	$2(0.1 \times 0.3) + 0.2^2 = 0.1$
25	$2(0.4 \times 0.1) + 2(0.2 \times 0.3) = 0.2$
30	$2(0.2 \times 0.4) + 0.3^2 = 0.25$
35	$2(0.3 \times 0.4) = 0.24$
40	$(0.4)^2 = 0.16$

Deriving Probabilities From Discrete Distributions

(9) Michael Dickson, who plays for the Seattle Seahawks, is a punter in the NFL. The following sampling distribution gives the mean punting distance (in yards) for a sample of $n = 4$ punts. Use this distribution to answer the following questions:

$\bar{X}$	Mean Punt Distance	$P(\bar{X})$
35.33		0.40
45.67		0.50
55.00		0.08
60.33		0.01
65.67		0.01

- What is the probability the mean distance is 55 or more yards?

$$\begin{aligned}
 P(\bar{X} \geq 55) &= 1 - P(\bar{X} < 55) \\
 &= 1 - (0.4 + 0.5) = 0.1
 \end{aligned}$$

- What is the probability the mean distance is at least 35.33 yards?

$$P(\bar{X} \geq 35.33) = 1$$

- What is the probability the mean distance is between 35.33 and 45.67 yards?

$$P(35.33 \leq \bar{X} \leq 45.67) = P(X = 35.33) + P(X = 45.67) = 0.9$$

- What is the probability the mean distance is 35.33 yards or 55 yards?

Let A be the event $\bar{X} = 35.33$ and B be the event $\bar{X} = 55$. These outcomes are disjoint, so

$$P(A \cup B) = P(\bar{X} = 35.33) + P(\bar{X} = 55) = 0.4 + 0.08 = 0.48$$

(10) A random variable X follows a binomial distribution with probability of success $p = 0.43$ and number of trials $n = 8$. What is the probability of observing fewer than 3 successes, that is what is $P(X < 3)$?

Let $X \sim \text{Binomial}(p = 0.43, n = 8)$ then

$$P(X < 3) = P(X = 0) + P(X = 1) + P(X = 2)$$

$$= \sum_{k=0}^2 \frac{8!}{k!(8-k)!} \times 0.43^k \times (1 - 0.43)^{8-k} \approx 0.256$$

(11) Tom Brady is a famous quarterback who recently retired from playing in the National Football League. During his career, the probability that Brady would complete a pass in a given play was 64.3%. If a typical NFL team runs about 35 offensive pass plays, what is the probability that Brady would complete exactly 28 pass plays in a game?

Let $X \sim \text{Binomial}(p = 0.643, n = 35)$ then

$$P(X = 28) = \frac{35!}{28!(35-28)!} \times 0.643^{28} \times (1 - 0.643)^{35-28} \approx 0.021$$

(12) A random variable W follows a Poisson distribution with rate parameter $\lambda = 1.8$ events per hour. What is the probability of observing exactly four events in a given hour, that is what is $P(W = 4)$?

Let $W \sim \text{Poisson}(\lambda = 1.8)$

$$P(W = 4) = \frac{1.8^4 \cdot e^{-1.8}}{4!} \approx 0.072$$

(13) A factory experiences on average 3 equipment malfunctions per day, following a Poisson distribution with a rate parameter of 3. Calculate the probability that on a randomly selected day, the factory will observe 3 or more equipment malfunctions.

let X represent the number of equipment failures in a given day for the factory, then

$$X \sim \text{Poisson}(\lambda = 3)$$

$$P(X \geq 3) = 1 - P(X < 3)$$

where

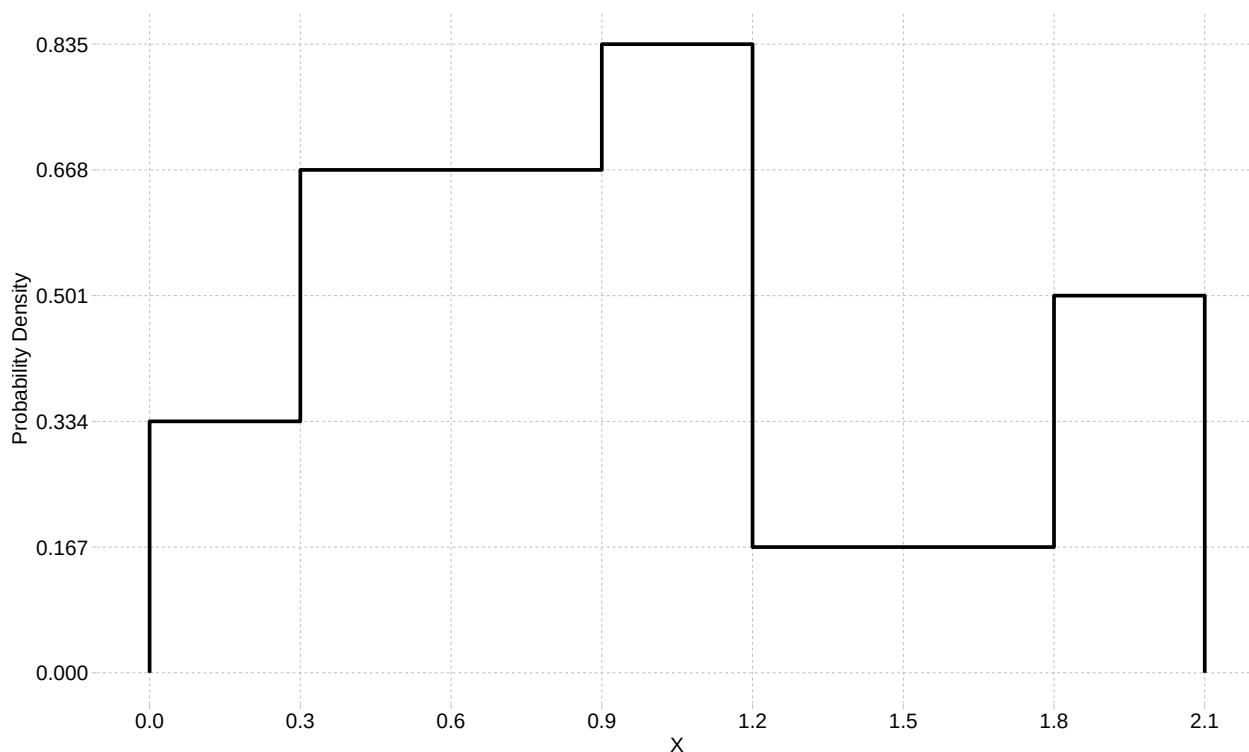
$$P(X < 3) = P(X = 0) + P(X = 1) + P(X = 2) = \sum_{k=0}^2 \frac{3^k \cdot e^{-3}}{k!} \approx 0.423$$

so

$$P(X \geq 3) = 1 - 0.423 \approx 0.577$$

Deriving Probabilities From Continuous Distributions

(14) Use the plot below of the probability distribution of a continuous random variable X to answer the following questions



Use the plot to find the following probabilities:

- What is $P(X < 0.6)$?

In this plot the area comprising a single grid square is

$$\text{Area of a rectangle} = \text{Base} \times \text{Height} = 0.167 \times 0.3 \approx 0.05$$

there are six grid squares between 0 and 0.6, therefore

$$P(X < 0.6) = 6 \times 0.05 = 0.3$$

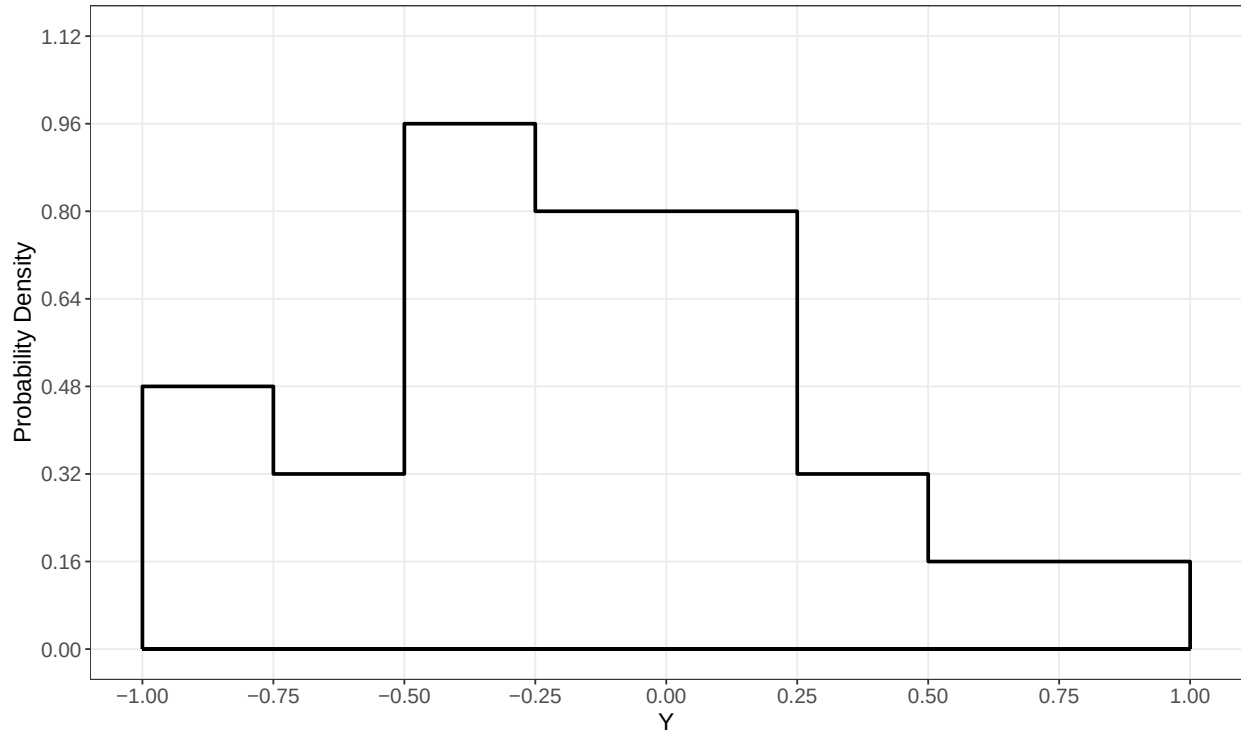
- What is $P(X > 0.3)$?

$$P(X > 0.3) = 1 - P(X \leq 0.3) = 1 - (2 \times 0.05) = 0.9$$

- What is $P(1.5 < X < 1.8)$?

$$P(1.5 < X < 1.8) = 0.05$$

(15) Use the plot below of the probability distribution of a continuous random variable Y to answer the following questions



Use the plot to find the following probabilities (recall that the area of a triangle is $\frac{1}{2} \times \text{base} \times \text{height}$)

- What is $P(Y > -0.25)$?

In this plot the area comprising a single grid square is

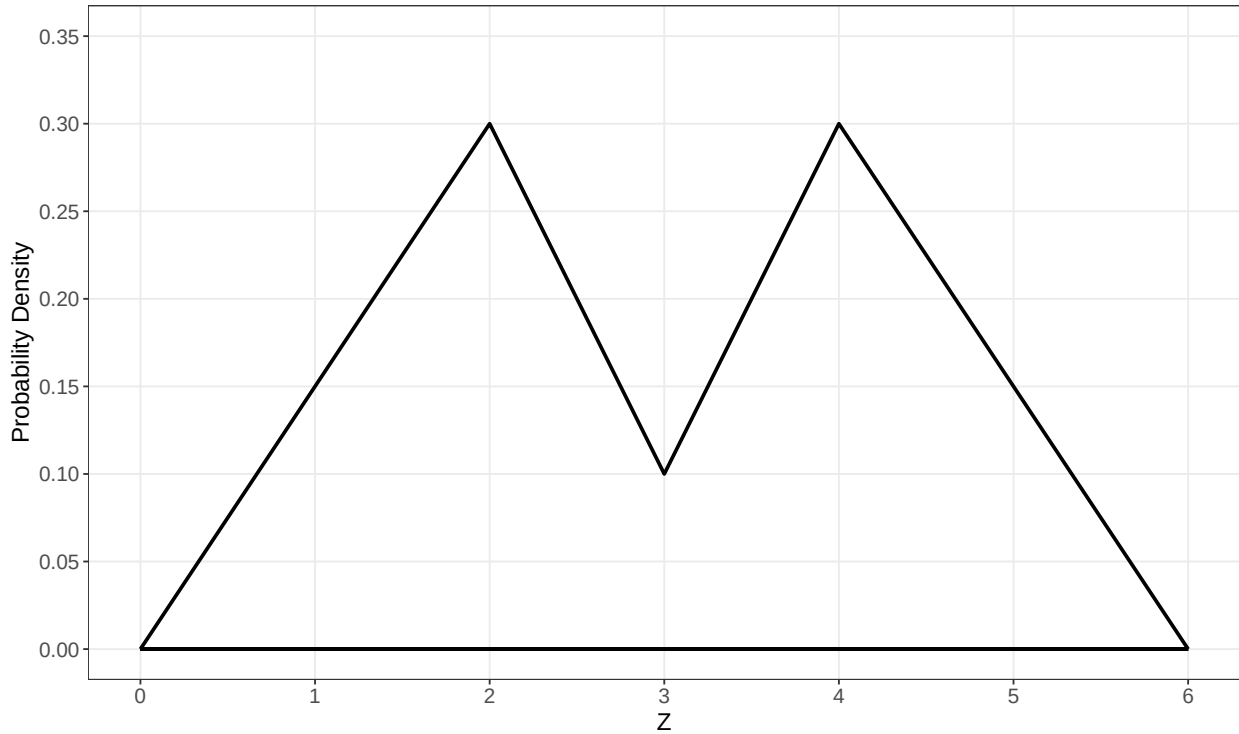
$$\text{Area of a rectangle} = \text{Base} \times \text{Height} = 0.25 \times 0.16 \approx 0.04$$

There are 14 grid squares in the area between -0.25 and 1.0 $P(Y > -0.25) = 14 \times 0.04 = 0.56$

- What is $P(-0.5 < Y < 0.25)$?

There are 16 grid squares in the area between -0.5 and 0.25 therefore $P(-0.5 < Y < 0.25) = 16 \times 0.04 = 0.64$

(16) Use the plot below of the probability distribution of a continuous random variable Z to answer the following questions



Use the plot to find the following probabilities (recall that the area of a triangle is $\frac{1}{2} \times \text{base} \times \text{height}$)

- What is $P(Z < 2)$?

$$P(Z < 2) = 0.5(2)(0.3) = 0.3$$

- What is $P(2 < Z < 3)$?

In this plot a single grid square has an area of 0.05 thus the area between $Z = 2$ and $Z = 3$ is given by $P(2 < Z < 3) = 2(0.05) + (0.5)(1)(0.2) = 0.2$

- What is $P(Z > 4)$?

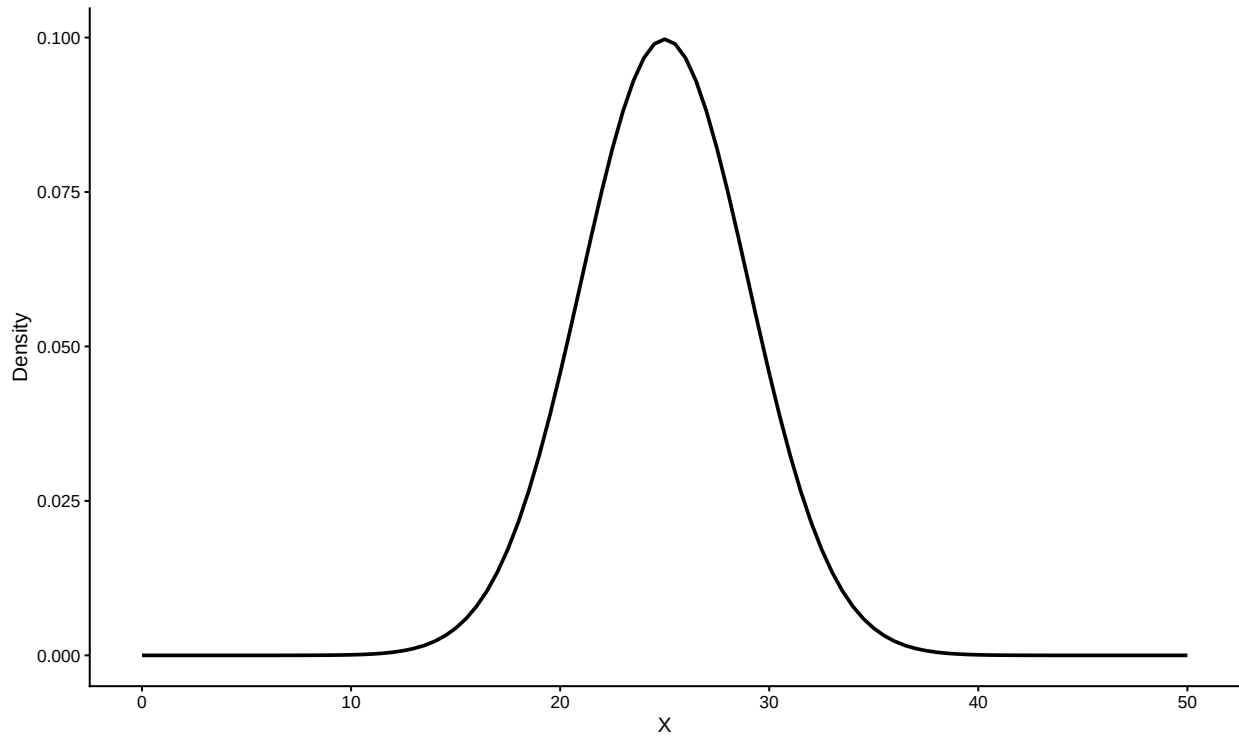
$$P(Z > 4) = P(Z < 2) = 0.3$$

- Give the approximate value of the 25th percentile of distribution of Z

Reading from the plot, the area to the left of $Z = 2$ is 0.3, so the 25th percentile lies just below 2. Working it out exactly: on $[0, 2]$ the density is the line $f(z) = 0.15z$, so the area to the left of z is $0.075z^2$. Setting $0.075z^2 = 0.25$ gives

$$Q1 = \sqrt{0.25/0.075} \approx 1.83$$

17.) Assume the continuous random variable X follows a normal distribution with mean $\mu = 25$ and variance $\sigma^2 = 16$.



Using a Z-table, find the following (approximate) probabilities for X . You can also use the app in the course website or StatDistributions.com to check your result

- What is $P(X < 15)$?

$$z = \frac{15 - 25}{4} = -2.5$$

$$P(Z < -2.5) = 0.006$$

- What is $P(X > 28)$?

$$z = \frac{28 - 25}{4} = 0.75$$

$$P(Z > 0.75) = 1 - P(Z \leq 0.75) = 1 - 0.773 = 0.227$$

- What is $P(12 < X < 20)$?

$$z_{\text{lower}} = \frac{12 - 25}{4} = -3.25$$

$$z_{\text{upper}} = \frac{20 - 25}{4} = -1.25$$

$$P(-3.25 < Z < -1.25) = P(Z < -1.25) - P(Z < -3.25) = 0.106 - 0.001 = 0.105$$

- What is $P(X > 30)$?

$$z = \frac{30 - 25}{4} = 1.25$$

$$P(Z > 1.25) = 1 - P(Z \leq 1.25) = 1 - 0.894 = 0.106$$

- What is $P(25 < X < 33)$?

$$z_{\text{lower}} = \frac{25 - 25}{4} = 0$$

$$z_{\text{upper}} = \frac{33 - 25}{4} = 2$$

$$P(0 < Z < 2) = P(Z < 2) - P(Z < 0) = 0.977 - 0.5 = 0.477$$

- What is the value of X that marks the 95th percentile of the distribution?

the 95th percentile of a standard normal distribution is $z = 1.645$ plugging this into the z -score equation and solving for X gives

$$(z \times \sigma) + \mu = (1.645 \times 4) + 25 = 31.58$$

- What is the value of X that marks the 40th percentile of the distribution?

the 40th percentile of a standard normal distribution is $z = -0.253$

$$(z \times \sigma) + \mu = (-0.253 \times 4) + 25 = 23.99 \approx 24.0$$

18.) Suppose that a random variable X follows a normal distribution with mean $\mu = -5$ and standard deviation $\sigma = 1.8$. Using the plot of the Z -distribution below, shade in the area corresponding to the probability $P(-2 < X < 0.5)$

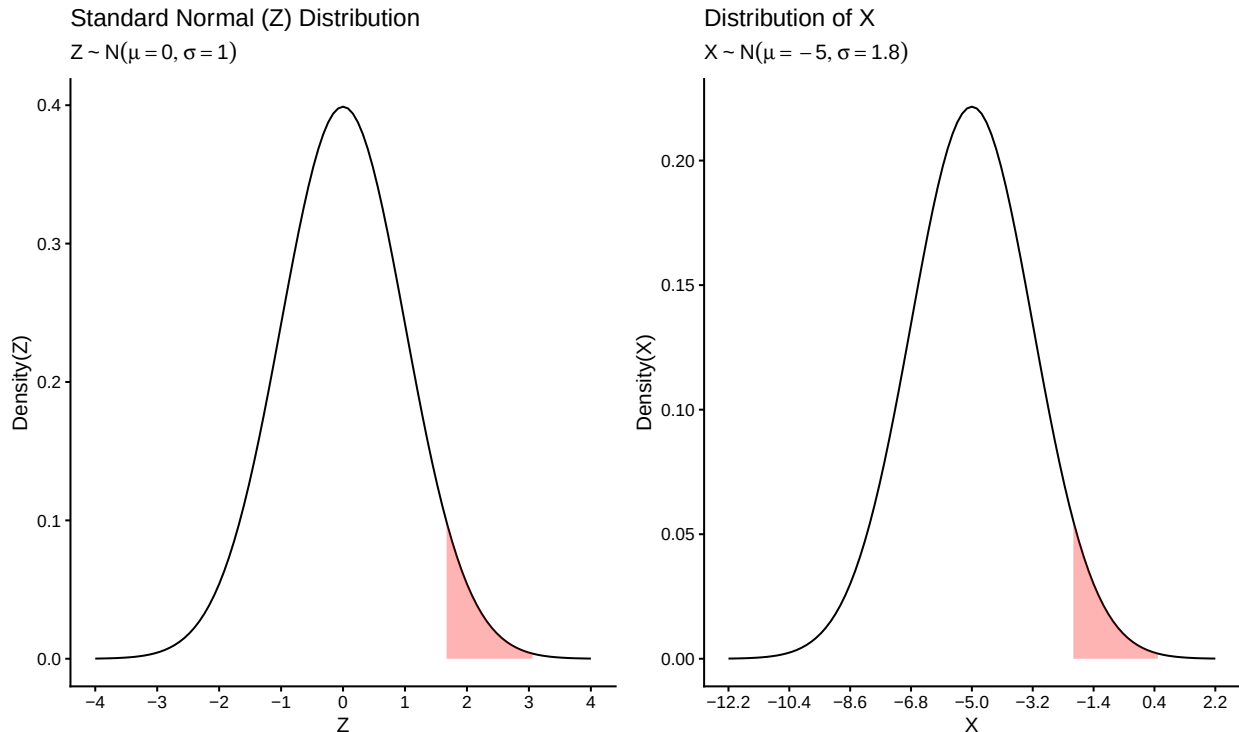
start by finding the z -scores

$$P(-2 < X < 0.5) = P(X < 0.5) - P(X < -2)$$

$$z_{\text{lower}} = \frac{-2 - (-5)}{1.8} = 1.67$$

$$z_{\text{upper}} = \frac{0.5 - (-5)}{1.8} = 3.06$$

$$P(-2 < X < 0.5) = P(1.67 < Z < 3.06)$$



References